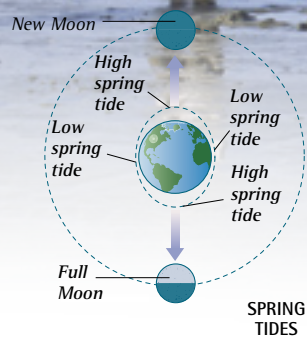
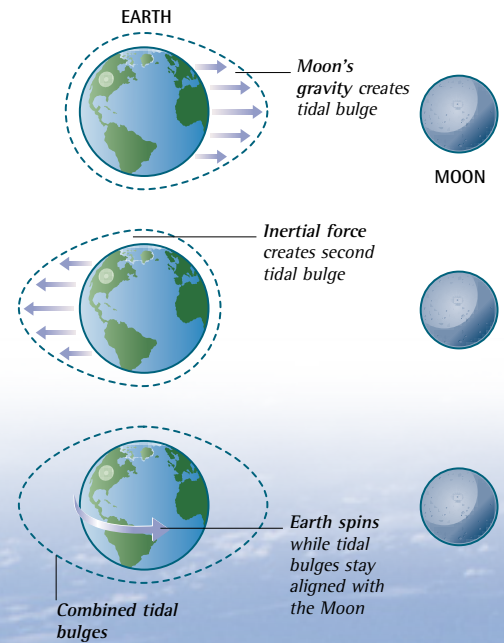


TIDES AND TIDAL STREAMS

Coasts and coastal waters in most parts of the world are strongly affected by the rhythm of the tides. These are caused by the gravity of the Moon constantly drawing ocean water toward it while the Earth spins. They are modified by other forces, including the gravity of the Sun. Tides are also influenced by the shape of the shoreline, which creates extreme tidal effects on some coasts, while others are almost tideless.

LUNAR ATTRACTION ►

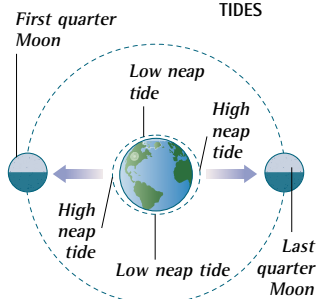
As the Moon orbits the Earth, its gravity attracts ocean water to create a tidal "bulge." But the Moon's movement also makes the Earth move in space, slinging water away from the Moon to create another tidal bulge. As the Earth spins, its coasts move in and out of the tidal bulges, causing high and low tides.



SPRING TIDES



NEAP TIDES

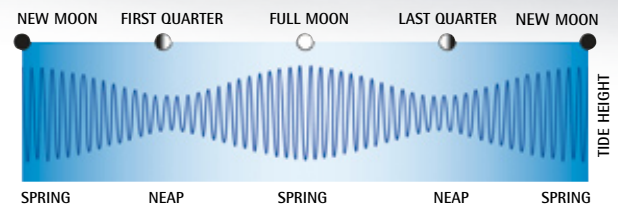


▲ DAILY CYCLES

If the entire planet were covered by oceans, the two tidal bulges would cause two equal high tides and low tides every day, with the time of high tide advancing by about 50 minutes every 24 hours as the position of the orbiting Moon changes. This is exactly what happens on many coasts. But the continents interfere with the smooth passage of the tidal bulges over the spinning Earth, giving some regions unequal tides. The shores of northern Vietnam, for example, experience just one tidal cycle a day.

◀ MONTHLY CYCLES

As the Moon orbits the Earth, it moves in line with the Sun twice a month at the times of full Moon and new Moon. When the Sun and Moon are aligned like this, their combined gravity causes extra-large spring tides every two weeks. In the weeks between a full and new Moon, the gravity of the Sun offsets that of the Moon, reducing the tidal bulge effect and causing less dramatic neap tides.



▲ SPRINGS AND NEAPS

The monthly cycle of spring tides and neap tides affects the total range of the tide, and not just its maximum height. A spring tide falls much lower than a neap tide, exposing more of the shore, as well as rising much higher a few hours later. The tidal range changes every day, increasing during the week before a full or new Moon, and decreasing as the phase of the Moon approaches its first or last quarter.

